dataset, since there is one “half” of it (where `x<0`) that has a sensibly lower density than the rest of the dataset (you can also see this from the reachability distances: the green area has a dramatically lower average for the reachability distance). This is where the hierarchical clustering’s added value comes into play; when we run the main OPTICS algorithm, we produce an ordering and a reachability plot: these doesn’t provide immediately a clustering, but they are (or rather imply) a set of flat clusterings, that we can obtain by “cutting” the reachability plot with a specific value of ε taken from `[0, εMAX]`.

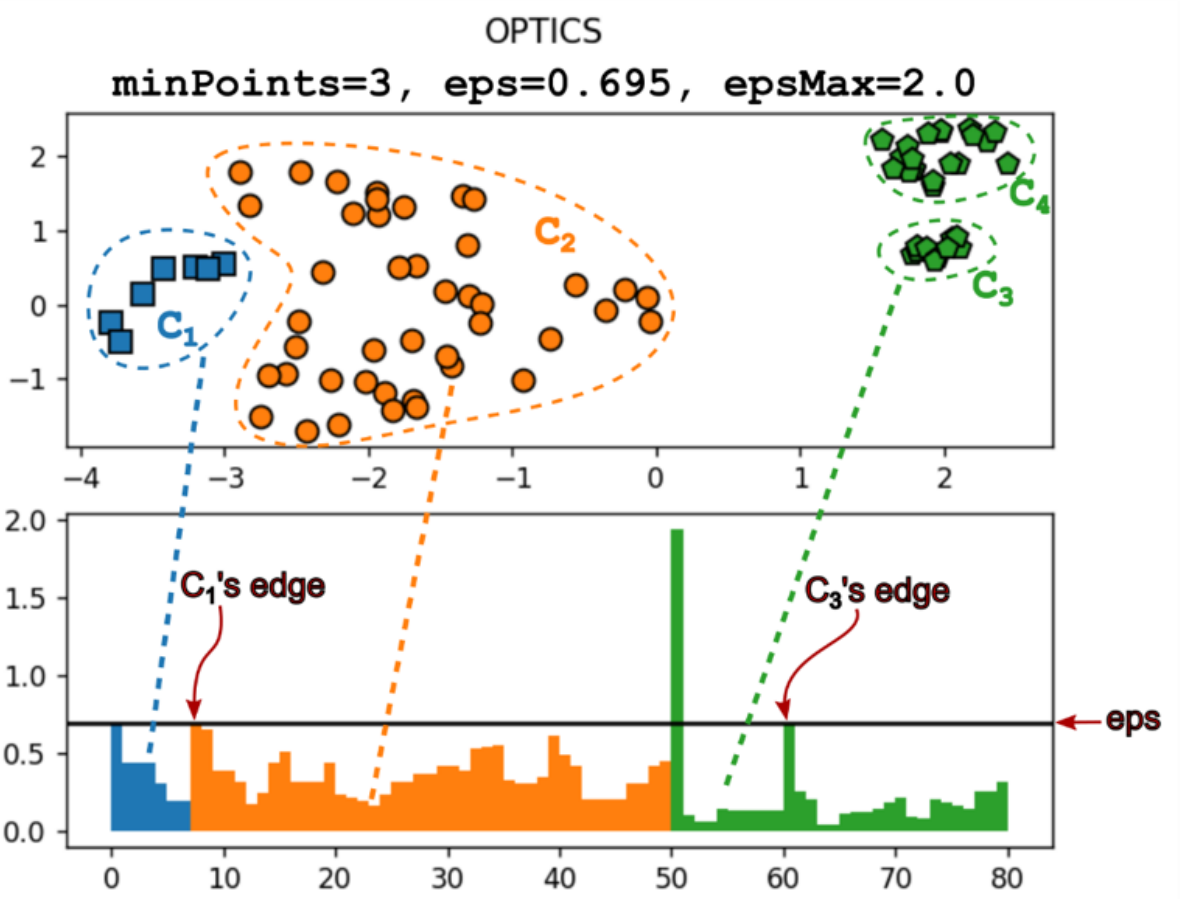


Figure 12.21 Yet another reachability plot with ε==0.695. This value is smaller than the reachability